

Reflection Journal

My Observations About the Iris Dataset

Dataset Overview:

- Number of samples: 150
- Number of features: 4
- Number of classes: 3

Key Findings from the Visualization

- The iris setosa cluster forms a distinct, tight group with shorter sepal lengths and wider sepal widths, making it easily distinguishable from the other two species.
- Iris versicolor and Iris virginica overlap significantly in sepal dimensions, particularly around sepal length of 5.5 -6.5cm and sepal widths of 2.5-3.0cm.
- Iris setosa features a higher sepal width relative to its length. Conversely, as sepal length increases across versicolor and virginica, sepal width stays relatively constant or increases at a much lower rate

Questions for Further Investigation

- Would plotting petal length vs petal width provide a clearer linear boundary between Iris versicolor and Iris virginica than sepal measurements?
- Which machine learning would achieve higher accuracy when predicting species using all four features combined?

Reflection

By completing this exercise, I learned how seamlessly pandas DataFrames structure raw arrays into labelled tables and how NumPy simplifies quick array computations like mean and standard deviation. Additionally, visualizing data using Matplotlib revealed natural groupings and feature overlaps that were not obvious from looking at numeric tables alone.